



USN

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

IS3242-1 – GRAPHICS AND ANIMATION
(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following is NOT one of the four major tasks in computer graphics?
A) Modeling B) Rendering
C) Animation D) Compression
2. Polygons in computer graphics are defined by:
A) Their color B) Their vertices
C) Their texture D) Their lighting
3. The primary purpose of antialiasing is to:
A) Enhance lighting effects B) Smooth out jagged edges
C) Increase polygon count D) Reduce rendering time
4. Display lists can improve performance by:
A) Reducing memory usage B) Pre-compiling graphics commands
C) Increasing complexity D) Running time
5. The primary function of the OpenGL API is to:
A) Manage databases B) Render 2D and 3D graphics
C) Process audio D) Handle text input
6. Which of the following is a common type of graphics output?
A) Raster graphics B) Vector graphics
C) Both A and B D) Vertex graphics
7. Which of the following is a technique for creating 3D objects?
A) Polygon modeling B) Image editing
C) Texturing D) Image acquisition
8. Which of the following techniques is used for hidden surface removal?
A) Z-buffering B) Texture mapping
C) Rasterization D) Modelling
9. Projections in OpenGL are defined by:
A) Matrices B) Vectors
C) Polygons D) Textures
10. The primary purpose of the Z-buffer is to:
A) Store color values B) Manage depth information
C) Create textures D) Manage Intensity Information

11. The main purpose of the window-to-viewport transformation is to:
 A) Change the size of the viewport B) Map coordinates from one space to another
 C) Adjust camera position D) Normalization
12. The parametric representation of an ellipse requires:
 A) Two radii B) A single radius
 C) Only integer values D) Four radii
13. Bezier curves are defined by:
 A) Control points B) Fixed equations
 C) Simple shapes D) Radius
14. The painter's algorithm works by:
 A) Sorting polygons by depth B) Rendering polygons in order of appearance
 C) Using a Z-buffer D) Using a A * Algorithm
15. The representation of quadric surfaces involves:
 A) Second-degree equations B) Linear equations
 C) Cubic equations D) Exponential equations
16. The positioning of the camera in computer graphics affects:
 A) The perspective of the scene B) The color of objects
 C) The size of objects D) The shear of the object
17. Key framing in animation helps to
 A) Define start and end points B) Create fluid motion
 C) Simplify character design D) Define movement of the frame
18. Digital image formats can include:
 A) BMP, GIF, TIFF B) TXT, DOC, PDF
 C) HTML, CSS, XML D) PDF, XSL, DOCX
19. Histogram equalization works by:
 A) Redistributing pixel intensities B) Adjusting brightness levels
 C) Smoothing images D) Shaping images
20. Median filtering is particularly effective for:
 A) Removing random noise B) Enhancing edges
 C) Increasing brightness D) Lowering the brightness

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe the OpenGL Graphic pipeline.	8	L*2	1	1
	b) Explain with equations the object as seen from a pinhole camera.	4	L2	1	1
	c) Write an Open GL program to display a circle.	4	L3	1	2
2.	a) Write the GL program for drawing a smiling face(reshape,keyboard and idel functions not needed)	8	L3	2	2
	b) Distinguish between immediate and retain mode of operation in display's.	4	L2	2	1
	c) Explain orthographic projections in OpenGL with gluOrtho2d() and an example.	4	L2	2	1
3.	a) Explain modelling, Geometry processing, rasterization and fragment processing.	8	L2	2	1
	b) Describe Scanline algorithm for rasterizing a polygon.	8	L2	2	1
Unit – II					
4.	a) Describe concatenation of transformation with an example, viz. rotation about a fixed point.	8	L2	3	1
	b) Describe window to viewport transformation.	8	L2	3	1
5.	a) Describe simple projections with an example.	8	L2	3	1

b)	Explain perspective and parallel Projections in Open GL with different gl functions.	8	L2	3	1
6. a)	Explain Area Subdivision method for visible surface detection with neat diagrams.	8	L2	3	1
b)	Explain the parametric representation of an ellipse.	8	L2	3	1
Unit – III					
7. a)	Explain the Phong lighting model.	8	L2	4	1
b)	Explain with an example Cohen Sutherland line clipping.	8	L2	4	1
8. a)	Explain physics based animation and procedural techniques of animation.	8	L2	5	1
b)	Explain any two image enhancement techniques.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
